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RESEARCH

Automatic exposure strategy network for robust visual odometry in environments with high dynamic range

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Abstract
The success of visual-based localization requires the image features to be sufficiently captured in well-exposed images and reliably tracked among image sequences. In this paper, a learning-based exposure strategy algorithm is developed to improve the performance of visual odometry or simultaneous localization and mapping in scenes with sudden illumination changes and high dynamic range (HDR). The exposure value is predicted by optimizing a novel image information metric, which is calculated based on a multi-scale image histogram. Experimental observations show that the proposed metric is highly related to the localization performance, and achieves better localization accuracy when compared to other state-of-the-art metrics. Using this novel metric, the exposure control network is designed to predict the exposure value for the next image given a sequence of recently captured images as inputs. For different application scenarios, alternative network frameworks are proposed with the options to include an optimization module providing improved exposure values for feedback learning, but with additional computation complexity. The approach is light-weight for real-time video applications and the images captured using the predicted exposure values are well-exposed with better detected and matched features in a variety of challenging HDR scenes. Code will be available in https://github.com/leejieyu13/HDR-AE.

Keywords Exposure control · High dynamic range · Visual odometry

1 Introduction

Vision-based localization and mapping methods have been widely studied in robotics communities. Factors like light-blocking and changes in view directions contribute to fluctuations in illumination, which poses substantial challenges to these methods. It is difficult to detect sufficient features in poorly exposed images or track features between two frames with large brightness differences. In these situations, the localization is inaccurate and may fail completely [11].

Some post-processing methods have been developed to enhance the robustness of visual localization in challenging lighting conditions, including tone-mapping algorithms [7, 10, 13] and brightness-invariant feature detection and matching [1, 4, 15]. However, these approaches can be useful only when enough information is originally captured in the images. Pre-acquisition methods, actively adjusting the camera exposure settings when capturing images, are therefore crucial but have not received much attention.

Design of auto-exposure (AE) algorithms primarily comprises two components: the control method and the metric used as the control target. For vision-based localization, the metric should effectively reflect the quantity and quality of structural information in images. While gradient distribution [2, 24, 26, 28, 29] and entropy [11, 12] can be promising indicators for image information, they are very sensitive to image noise. The number of feature matches in sequential image pairs was used as the optimized metric in [27]. However, [2] revealed that the cross-correlation between the count of good matches and the pose estimation error is nearly negligible. The reason is that image data is highly redundant, and the benefit of simply using more data points quickly flattens off. The experiments in [5] demonstrated that a better localization performance can be achieved by

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using more equally distributed points across space, even if some weaker gradient points are included. This indicates that the spatial distribution of detected features is an important factor for visual localization.

For control methods, most of the existing work focuses on photometric calibration to predict the exposure adjustment effects and the iterative algorithms to search for optimal control [2, 3, 28, 29]. There are limitations to these approaches. Firstly, it is challenging to accurately predict the effects of exposure adjustment on the saturated regions due to clipped information in standard dynamic range (SDR) images. Moreover, iterative optimization might not be feasible for real-time localization in terms of processing time. More recently, deep convolutional neural networks [20, 27] have been developed to proactively adjust exposure settings to compensate for impending light changes.

In this paper, a novel image information metric and an exposure control network are proposed. The metric evaluates the cosine similarity between multi-scale image histograms and the uniform histograms. It is observed that the more evenly distributed the histogram is on the full range of intensities, the better feature detection the image can gain. By optimizing the proposed metric, the resulting image exhibits a broader distribution of features across space. With the proposed metric, the exposure control network leverages a sequence of recent images and their corresponding exposure settings to learn the future irradiance information, and predicts the optimized exposure settings that can be applied before the next image is captured. To further improve the prediction accuracy, several iterations of Nelder–Mead (NM) optimization [18] can be run based on the captured image and the improved exposure value is fed back to the network for feedback learning.

In summary, the main contributions are:

- A novel multi-scale histogram-based image information metric is designed for loss function calculation, which is strongly related to localization performance. Experiments show that the proposed metric outperforms other state-of-the-art metrics in terms of localization accuracy and noise sensitivity.
- An exposure strategy network is developed to predict the optimized exposure value for the next image by leveraging a sequence of recent images. A joint learning scheme is used to obtain a more representative latent feature. Alternative network frameworks are proposed with the options to include an optimization module providing fine-tune exposure value for feedback learning. The proposed system can quickly converge and capture well-exposed images for visual odometry in a variety of challenging high dynamic range scenes.
- A new benchmark dataset is also provided which includes HDR RAW image sequences in changing lighting conditions and high dynamic range scenes, to facilitate future studies of exposure control, which can be found in https://drive.google.com/drive/folders/1aTjSEuPMKvv19RZQatH6E0kyJoQeUn8I.

2 Related work

2.1 Optimization metrics

Exposure control algorithms for robot vision typically aim for better feature detection and tracking. Since keypoints with strong gradients are often used as good features to track [23], some methods adjust the exposure settings to maximize gradient-based metrics. In order to minimize sensitivity to noise, weak gradients need to be filtered or weighted properly. A logarithm mapping function was proposed in [24, 25], which is activated after a certain gradient level, to better balance the importance of small and large gradients. Similarly, Zhang et al. [29] used the percentile of all the gradient magnitudes as the robust metric, and introduced the soft percentile as a smooth and differentiable approximation for optimization. These methods tend to embrace more saturation in order to increase the overall gradient. To mitigate this issue, Wang et al. [28] assigned greater weights to pixels that are more responsive to photometric variations. An entropy weighted gradient and a saturation based activation function were introduced in [12] to obtain well-exposed images with more image information. Shin et al. [26] further incorporated the metrics based on entropy, gradient and noise for control of exposure time and gain. In general, increasing the exposure time can reduce noise, but will lead to motion blur. This relationship typically means that the gain/exposure time balance must be considered together for visual odometry.

In [27], the exposure strategy maximizes the number of detected features and the number of matched inliers across frames. Nevertheless, the cross-correlation analyses in [2] shows that the number of good stereo matches does not correlate to pose estimation error. Onzon et al. [20] developed an exposure control network for object detection in HDR scenes, but the localization performance of the system was not evaluated.

In this paper, a novel image information metric is proposed based on a multi-scale histograms. When the intensities are more evenly distributed on the histogram utilizing the full range of intensities, better contrast and more image information can be obtained. While this can be measured by the cosine similarity between the image histogram and

the uniform histogram, such global measures are not effective in ensuring that feature points are widely distributed in space. In order to obtain better spatial distribution of feature points, the image information in non-overlapping blocks is evaluated, and the metric is averaged over all the blocks. This metric is highly related to the localization performance and leads to better localization accuracy in the comparison to other state-of-the-art metrics. Additionally, as shown in the experiments, the proposed metric is less sensitive to noise when compared to the gradient-based metrics or the entropy-based metrics.

2.2 Prediction models and control methods

Given the metrics, most existing exposure control methods use iterative optimization algorithms to find the optimal exposure settings, which requires multiple image acquisitions or simulation to sparsely sample the parameter spaces.

To predict the effects of exposure adjustment on the image metric, a series of synthetic exposure changes was simulated by applying various levels of gamma correction on a captured images in [25]. Some work [3, 29] further improved the prediction model based on the reconstruction of scene irradiance using camera response function. These systems are too slow to achieve real-time performance. In order to accelerate the optimization process, Wang et al. [28] developed a coarse-to-fine strategy to adjust the camera exposure settings.

Due to image saturation, the precision of irradiance reconstruction is compromised in HDR scenes, leading to challenges in accurate prediction on exposure adjustment effects. To compensate for the reconstruction error, a feedback on the difference between the actual and predicted gradient information metrics was incorporated in [2]. Model-free optimization methods, such as Bayesian optimization [12] and the Nelder-Mead (NM) method [26], were implemented in exposure control. However, without prediction models, capturing a sequence of images with varying exposures is required to make inferences.

Besides long processing time and inaccurate prediction, another main issue of these prior methods is the lack of capability to anticipate changes in lighting conditions. In order to proactively adjust the exposure settings, Tomasi et al. [27] proposed a learning-based exposure control network. The exposure settings are predicted by leveraging recent images and applied before the next image is captured. Without preview image acquisition and optimization iterations, neural networks can be a more practical solution for video applications with high frame rate.

Similar to [27], a network is developed in this paper to predict the exposure value with a sequence of recently captured images and the corresponding exposure settings as inputs. A few NM iterations (8 compared to the 25 suggested in [26]) can be implemented on the captured image, and the improved exposure value is obtained as a reference target for feedback learning. Moreover, in order to learn a more representative latent feature, a joint learning framework is designed with a branch predicting histogram of next frame.

3 Problem statement

In order to clearly state the problem, the relationship between the radiant power from the scene and the SDR image is first modeled. Following [9], the radiant power, ϕ, is defined in units of electrons per second (with its relationship to absolute scene irradiance not of importance). ϕt electrons are collected over t seconds, and converted to voltage. The voltage is amplified by camera gain g, and converted to a digital number as a pixel value. The product of exposure time t and camera gain g is known as exposure value EV. For a 12-bit RAW image, the pixels with intensity larger than $I_{\max} = 2^{12} - 1$ are clipped due to saturation. According to this process with noise incorporated, the RAW measurement for each pixel can be described by

$$I_{\mathrm{RAW}} = \min\{\lfloor g(\phi t + n_{\mathrm{pre}}) + n_{\mathrm{post}} \rfloor, I_{\max}\}, \tag{1}$$

where $\lfloor \cdot \rfloor$ represents quantization. n_{pre} is the noise introduced before amplification, including the noise from photon arrivals and from sensor readout. n_{post} is the noise after amplification, mainly from ADC.The RAW image is then fed into image signal processor (ISP), processed by demosaicing, white balance, gamma correction, etc., to generate an 8-bit SDR image. $\boldsymbol{I} = f(\boldsymbol{I}_{\mathrm{RAW}})$, where $f(\cdot)$ is a nonlinear and monotonic function, that maps increasing radiant power to increasing pixel values. It is modeled by a differentiable function for training in the proposed system.

In summary, an 8-bit SDR image $\boldsymbol{I}$ has limited dynamic range clipped by the saturation point and nonlinearity introduced by the ISP. The problem addressed in this paper is to select the optimal exposure value EV to obtain image $\boldsymbol{I}$ with good quantity and quality of structural information for robust visual-based localization in challenging HDR scenarios.

4 Proposed system

Details of the proposed system are introduced in the following sections, including feature extraction modules, network architecture and metric design. The overview of the proposed system can be found in Fig. 1. Given the exposure value, the image acquisition and feature extraction module

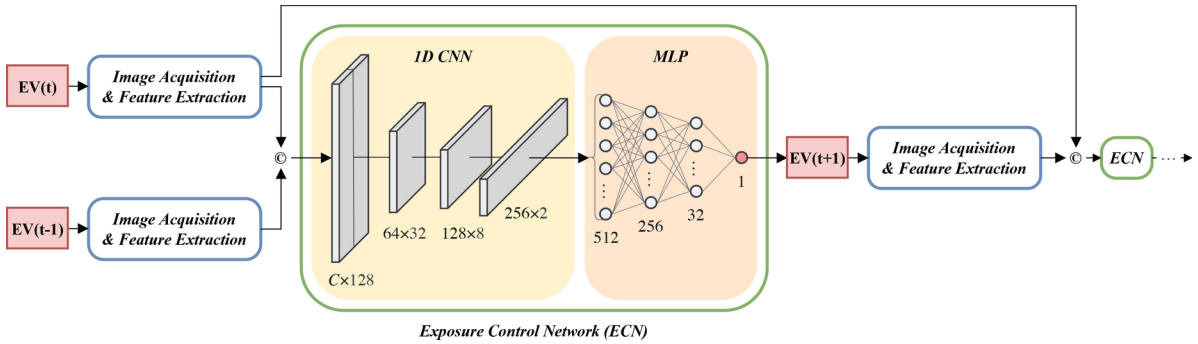


Fig. 1 Overview of the proposed system. The exposure control network takes features from two recently captured images as input, and predict the next exposure value

captures the image and generates input features. The features of two recently captured images are then concatenated and fed into the exposure control network to predict the next exposure value.

4.1 Image acquisition and feature extraction module

The network input is generated based on the histogram statistics and the corresponding exposure settings of the recently captured SDR image sequence $\{I(t), I(t-1)\}$. For different application scenarios, three alternatives of the image acquisition and feature extraction module are proposed, offering optional feedback learning to enhance accuracy while introducing additional computation complexity.

4.1.1 Multi-scale histogram

In order to use the image information locally and globally, histograms at three different scales are computed. For the coarsest scale, a 128-bin histogram is computed on the whole image based on the luma channel. Similarly, at the intermediate scale and the finest scale, the image is divided respectively into 2-by-2 and 5-by-5 non-overlapped sub-images and histograms are computed from every sub-image.

4.1.2 Feedback learning

Given a captured SDR image $\boldsymbol{I}$, the power radiant $\hat{\boldsymbol{\phi}}$ can be estimated by the reverse camera response function, but clipped by $I_{\max}/EV$ due to saturation. With $\hat{\boldsymbol{\phi}}$, the metric values at candidate EV can be evaluated. Thus, the exposure value can be further optimized. This improved exposure value EV_{NM} can be fed back into the network as input for feedback learning [8]. The network is then trained based on the feedback to minimize the loss function for future inputs.

The optimization is solved by the Nelder–Mead (NM) algorithm [18] in the proposed system. It does not require any derivative of the loss function, so the loss can be calculated based on the standard histogram, which is less time-consuming compared to the differentiable histogram approximation [21]. Typically, with sudden changes in lighting conditions, the preview SDR image might have large saturation regions and models therefore fail to predict the effect of exposure adjustment, which yields sub-optimal solutions. However, with the network offering a good initial exposure near the optimal value for acquiring the preview image, the image is well-exposed, and an improved value can be achieved by several iterations.

4.1.3 Module alternatives

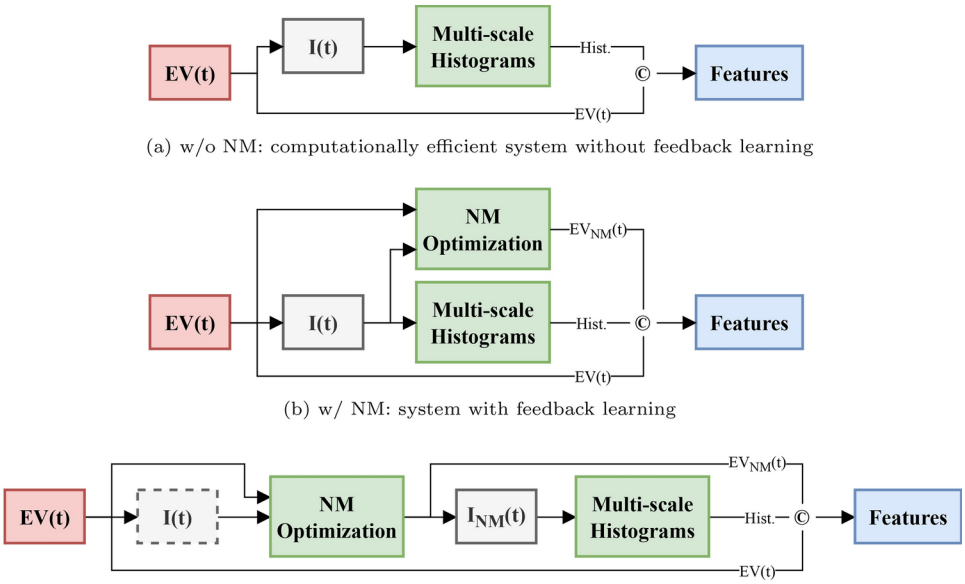
For the proposed system, three alternative image acquisition and feature extraction module are introduced as shown in Fig. 2, where (b) and (c) offer the choice to integrate the feedback learning.

For the first framework shown in Fig. 2a, the input feature contains the multi-scale histograms of the last two captured images $\{I(t), I(t-1)\}$ and the corresponding exposure values $\{EV(t), EV(t-1)\}$. Without optimization iterations in inference stage, this system has high computational efficiency.

Based on the first framework, a second framework is proposed to further improve the prediction accuracy by feedback learning, as shown in Fig. 2b. The reference exposure value for feedback learning is obtained by the NM optimization based on the captured image. The input feature becomes the concatenation of the multi-scale histograms, the corresponding exposure values $\{EV(t), EV(t-1)\}$ and the improved exposure values $\{EV_{\mathrm{NM}}(t), EV_{\mathrm{NM}}(t-1)\}$.

Alternatively, after the NM optimization, an image can be recaptured with the refined exposure value. As shown in Fig. 2c, the third framework can be considered as an optimization-based method with initialization given by the network outputs, which gives high accuracy for the exposure value prediction.

The comparison between the prediction accuracy and the required processing time of these three proposed



(a) w/o NM: computationally efficient system without feedback learning

(b) w/ NM: system with feedback learning

(c) w/ NM + recapture: precise prediction system with feedback learning and image re-acquisition. The resulting image $\boldsymbol{I}_{\mathrm{NM}}$ is re-captured with the NM-refined exposure value.

Fig. 2 Three alternatives for image acquisition and feature extraction module

frameworks can be found in the experiments. The selection of these frameworks depends on application scenarios, with the aim of balancing computational efficiency and accuracy.

4.2 Exposure control network

The auto-exposure control network consists three 1D convolutional layers with kernel size 4 and stride 4, followed by the multilayer perceptron (MLP) network containing three fully connected layers. All layers, except the output, use ReLU as the activation function.

Since exposure value is typically adjusted in stops that follow a power-of-2 progression, the output of the last layer is fed into a function that predicts binary logarithm of the exposure value,

$$\log_2(EV) = \mathrm{sigmoid}(x) \cdot (EV_{\max} - EV_{\min}) + EV_{\min}. \tag{2}$$

Given a sigmoid function in the range of 0 to 1, Eq. 2 maps the output of the last network layer x to exposure value $EV \in [2^{-8}, 2^{12}]$ with $EV_{\max} = 12, EV_{\min} = -8$.4.3 Image information metric

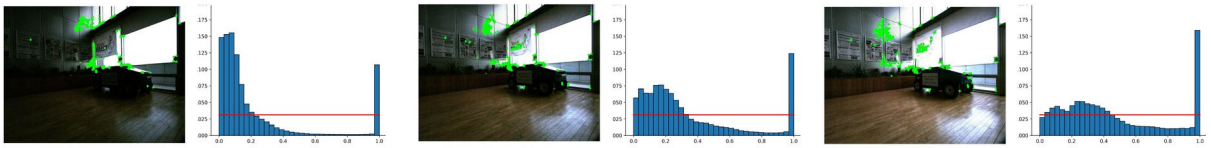
An image has better contrast and more useful information if its intensities are better distributed on the histogram utilizing the full range of intensities evenly, so image information could be evaluated by the cosine similarity between its histogram and a uniformly distributed histogram.

$$S_{\mathrm{c}}(\boldsymbol{h}) = \frac{\boldsymbol{h}_{\mathrm{ref}} \cdot \boldsymbol{h}}{\|\boldsymbol{h}_{\mathrm{ref}}\| \|\boldsymbol{h}\|}, \tag{3}$$

where $\boldsymbol{h}$ is the image histogram with N bins, and $\boldsymbol{h}_{\mathrm{ref}}(i) = 1/N, \ \ i = (1, 2, ..., N)$. N is set to be 128 in the experiments. For network training, [21] provides an approximate calculation method for differentiable histogram.Since spatial distribution of the features points is an important factor for localization task, in order to obtain more equally distributing features in space, the image information is also evaluated locally on a set of non-overlapping sub-images. The multi-scale image information metric is defined as

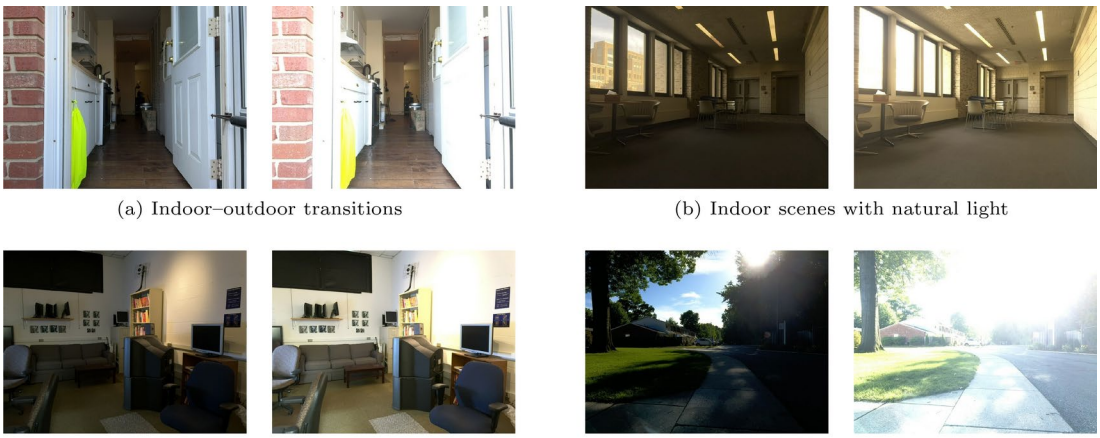
$$\begin{aligned} M &= w_{\mathrm{g}} M_{\mathrm{global}} + w_{\mathrm{l}} M_{\mathrm{local}} \\ &= w_{\mathrm{g}} S_{\mathrm{c}}[\boldsymbol{h}(\boldsymbol{\phi}, EV)] + w_{\mathrm{l}} \frac{1}{n} \sum_{i=1}^{n} S_{\mathrm{c}}[\boldsymbol{h}_i(\boldsymbol{\phi}, EV)], \end{aligned} \tag{4}$$

where $\boldsymbol{h}_{\mathrm{i}}$ $(i = 1, 2, ..., 9)$ are the histograms of the 3×3 non-overlapping sub-images. The weights for the global and local information metrics are set as $w_{\mathrm{g}} = 0.2$ and $w_{\mathrm{l}} = 0.8$. The loss function in the proposed system is the negative proposed metric.A visual example of differently exposed images is shown in Fig. 3, with the corresponding global and multi-scale image information metrics reported. By maximizing the global information metric M_{global}, Fig. 3b is selected. The relatively high contrast and less saturation regions in Fig. 3b shows that the cosine similarity between the image histogram and the uniform histogram can serve as a reliable indicator of image information. By the incorporation of local statistic in the multi-scale information metrics M, Fig. 3c is selected, which results in a more widely distributed set of detected features across the space.



$(a) M_{\text{global}} = 0.475,\ M = 0.392$ $(b) M_{\text{global}} = 0.563,\ M = 0.489$ $(c) M_{\text{global}} = 0.531,\ M = 0.517$

Fig. 3 Examples of different exposure settings. The histograms (32 bins for visualization) are shown with red line indicating the uniform histogram. Figure **(b)** with relatively high contrast and less saturation regions is selected by maximizing the global information matric M_{global}, while Figure **(c)** with more widely distributed ORB features (green dots) is selected by maximizing the multi-scale information metric M



(a) Indoor–outdoor transitions (b) Indoor scenes with natural light

(c) Indoor scenes with artificial light (d) Outdoor with tree shadows

Fig. 4 Examples of image sequences of different scenes. The images on the left and right are captured with different exposure settings respectively, showing the high dynamic range of the scenes

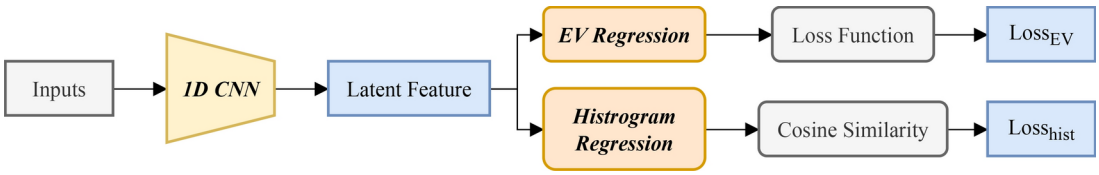


Fig. 5 Training framework. The exposure value branch and histogram branch are trained with the 1D CNN encoder iteratively to learn a more representative latent feature

5 Training

5.1 Dataset

In order to train and test the exposure control network, we introduce a dataset of linear HDR image sequences, including indoor scenes with artificial light or natural light from windows, outdoor scenes with shadows and indoor–outdoor transient environments. The image is generated by fusing 3 SDR RAW images, processed by denoising and alignment algorithms. Some examples are shown in Fig. 4. To perform the training, validation, and testing of the model, image sequences are generated by sampling the captured image sequence in both forward and backward directions, with intervals of one or two frames. Specifically, 80 sets of 20-frame sequences and 12 sets of 20-frame sequences are used for training and validation respective, while 6 sets of 45-frame sequences are used for testing.

5.2 Joint learning

With the HDR dataset simulating the real-world radiant power, SDR images captured with given exposure settings can be generated by the image formation model. The loss is calculated based on the simulated images to train the network. Inspired by the intuition that the exposure strategy is dependent on the scene irradiance, a joint learning framework, as shown in Fig. 5, is implemented in order to learn a more representative latent feature. The latent feature, as the output of the 1D CNN, is fed into two branches. The exposure value branch is trained to maximize the proposed image information metric in Eq. 4, while the histogram branch is trained to maximize the cosine similarity between

the 128-bin histogram prediction and the ground truth histogram on $\boldsymbol{I}(t+1)$. These two branches, including the 1D CNN, are trained iteratively with a learning rate as 10^{-4} and a batch size as 4 by Adam algorithm [14]. Experiments in Sect. 6.2.3 shows that the joint learning framework can efficiently improve the accuracy of exposure value prediction. In the inference stage, only the exposure value branch is performed.

6 Experiments

In this section, experiments are conducted to evaluate the proposed metric and the exposure control network.

6.1 Metric evaluation

6.1.1 Pose estimation

In this experiment, image pairs are selected using different state-of-the-art metrics [12, 25–29], and the pose estimation is evaluated.

The benchmark dataset in [26] is used. The images, with resolution 1600×1200, are taken by a stereo camera system with baseline approximately 20 cm. The intrinsic matrix, distortion parameters and relative pose are calibrated as ground truth, allowing a re-projection error of 0.097 pixel. The dataset consists 15 outdoor scenes and 10 indoor scenes with various illumination levels. For indoor scenes, the exposure time ranges from 4 ms to 67 ms with 3 ms intervals, and the gain ranges from 0 dB to 24 dB with step size 1 dB. For outdoor scenes, the exposure time ranges from 0.1 ms to 7.45 ms with 0.15 ms as intervals, and the gain ranges from 0 dB to 20 dB with step size 2 dB. Therefore for each scene, there are 550 stereo image pairs with different brightness and noise characteristics.

The optimal images are selected from the dataset according to different metrics, and then the relative pose between the stereo cameras is estimated based on the selected, undistorted image pairs. For pose estimation, two methods for feature detection and matching are implemented: ORB [22] and optical flow [16] with Shi-Tomasi features [23], with each method limited to a maximum of 1000 features. The essential matrix is found by the 5-point algorithm [19], and the relative pose is recovered. To deal with outliers, least median of squares (LMedS) and RANSAC algorithm (probability of success = 0.999, threshold = 0.001) are implemented. The rotation and translation error are calculated by

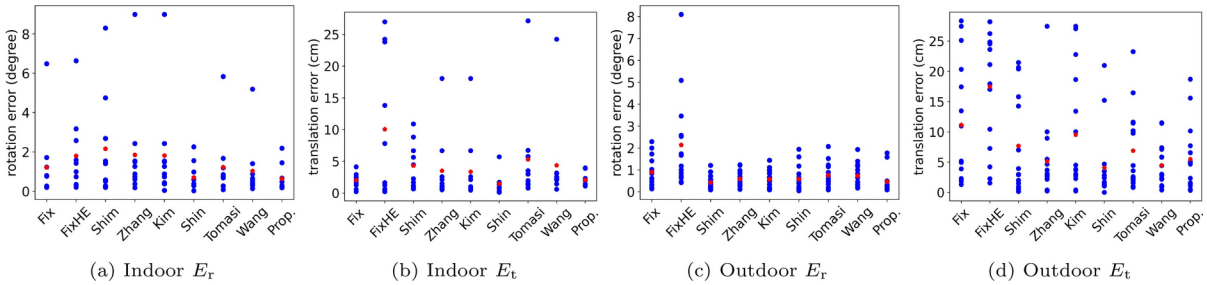
$$\begin{aligned} e_{\mathrm{r}} &= \arccos\{[\mathrm{tr}(\boldsymbol{R}_{\mathrm{GT}} \cdot \boldsymbol{R}^{\mathrm{T}}) - 1]/2\}, \\ e_{\mathrm{t}} &= \|\boldsymbol{T}_{\mathrm{GT}} - \boldsymbol{T}\|_2, \end{aligned} \tag{5}$$

where $\boldsymbol{R}_{\mathrm{GT}}$ and $\boldsymbol{T}_{\mathrm{GT}}$ are the ground truth rotation and translation, while $\boldsymbol{R}$ and $\boldsymbol{T}$ are the estimated rotation and translation.The comparison on pose estimation performance is shown in Table 1. Furthermore, take pose estimation based on ORB feature (ORB-LMedS) as an example, the errors for each indoor/outdoor scenes are plotted in Fig. 6. The result shows that the proposed metric achieves relatively strong average results and demonstrates robust performance, with a small worst-case error across various widely used pose estimation methods.

In addition to the state-of-the-art metrics, images captured with a fixed, relatively short exposure time and processed using histogram equalization are also evaluated, denoted as "Fix" and "FixHE" in the results. For indoor scenes, the settings are gain = 24 dB and exposure time = 4 ms, whereas for outdoor scenes, the settings are gain = 20 dB and exposure time = 0.25 ms. Ideally, a short exposure time helps to prevent motion blur and reduce delay. Histogram equalization can enhance contrast and gradients, making feature detection more effective. However, the results indicate that short exposure times increase noise, which can complicate pose estimation by reducing the reliability of feature detection. Underexposure can result in a loss of detail in darker regions of the image, making it harder to extract meaningful

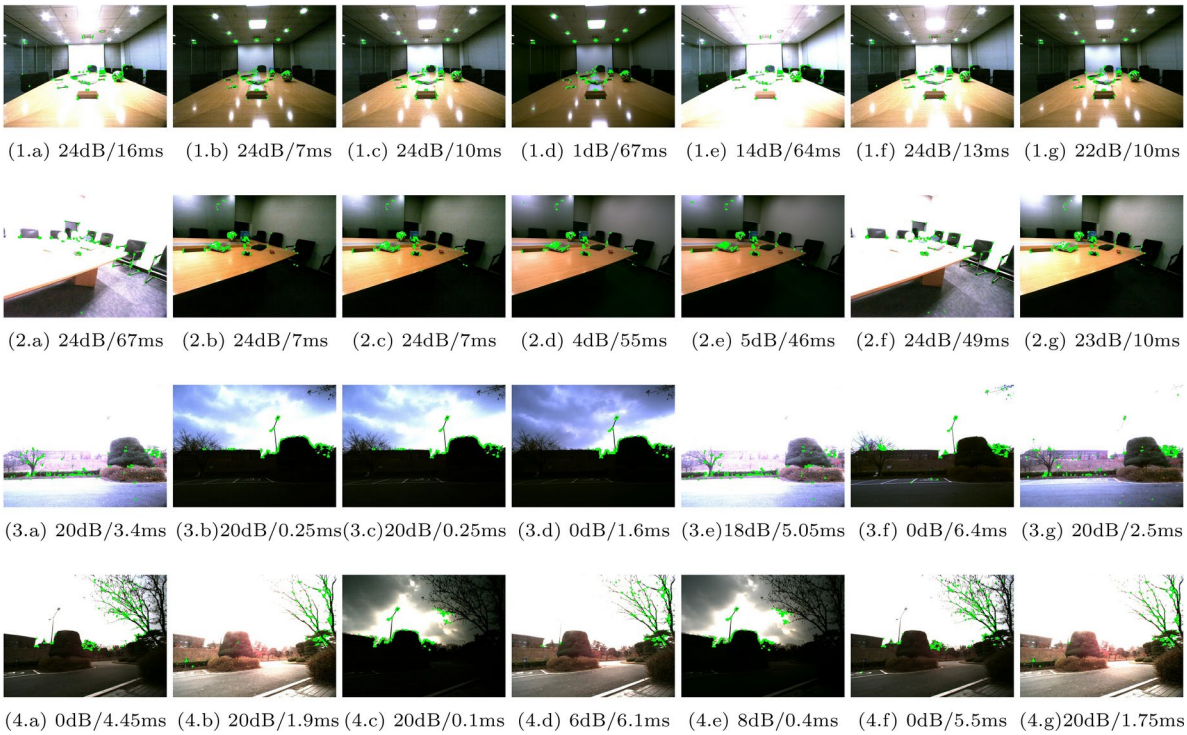
Table 1 Comparison on pose estimation, including average \ worst error (Bold: best; Italic: runner-up.)

	ORB-LMedS		ORB-RANSAC		OF-LMedS		OF-RANSAC	
	E_{r} (°)	E_{t} (cm)	E_{r} (°)	E_{t} (cm)	E_{r} (°)	E_{t} (cm)	E_{r} (°)	E_{t} (cm)
Fix	1.0 \ 6.5	7.5 \ 28.4	0.9 \ 6.5	5.8 \ 22.5	1.4 \ 6.9	9.1 \ 28.1	1.1 \ 4.4	5.9 \ 27.9
FixHE	2.0 \ 8.1	14.5 \ 28.2	1.3 \ 3.5	8.8 \ 28.2	1.0 \ **3.3**	6.7 \ 26.2	0.9 \ **2.7**	5.3 \ 27.3
Shim [25]	1.1 \ 8.3	6.3 \ 21.5	1.6 \ 30.3	4.3 \ 20.7	1.4 \ 8.4	5.6 \ 22.3	1.3 \ 7.6	5.3 \ 25.3
Zhang [29]	1.1 \ 9.0	4.5 \ 27.5	0.8 \ 9.0	*2.4* \ **17.0**	1.1 \ 6.6	5.5 \ 25.8	0.8 \ 3.7	4.5 \ 27.8
Kim [12]	1.1 \ 9.0	7.1 \ 27.5	0.8 \ 9.0	3.9 \ 27.1	1.2 \ 6.6	7.6 \ 25.8	0.8 \ 3.7	5.7 \ 27.8
Shin [26]	*0.6* \ *2.3*	**3.0** \ *21.0*	*0.6* \ *2.8*	**2.2** \ 21.0	1.1 \ 9.6	5.2 \ 21.4	0.9 \ 8.0	4.2 \ 25.6
Tomasi [27]	1.0 \ 5.8	6.3 \ 27.1	0.8 \ 3.4	6.0 \ 27.4	**0.8** \ *5.0*	5.6 \ 21.8	*0.6* \ 4.6	4.2 \ 21.8
Wang [28]	0.9 \ 5.2	4.4 \ 24.2	0.7 \ 5.2	3.5 \ 24.2	1.0 \ 9.3	**3.2** \ *14.6*	*0.6* \ 3.8	**2.7** \ **14.6**
Proposed	**0.5** \ **2.2**	*4.1* \ **18.7**	**0.5** \ **2.7**	3.7 \ *18.7*	*0.9* \ 5.3	*3.8* \ **12.9**	**0.5** \ *3.1*	*3.4* \ *14.8*



(a) Indoor E_r (b) Indoor E_t (c) Outdoor E_r (d) Outdoor E_t

Fig. 6 ORB-LMedS pose estimation errors for indoor and outdoor scenes (blue dots are the errors for each scenes and red dots are the average errors)



(1.a) 24dB/16ms (1.b) 24dB/7ms (1.c) 24dB/10ms (1.d) 1dB/67ms (1.e) 14dB/64ms (1.f) 24dB/13ms (1.g) 22dB/10ms

(2.a) 24dB/67ms (2.b) 24dB/7ms (2.c) 24dB/7ms (2.d) 4dB/55ms (2.e) 5dB/46ms (2.f) 24dB/49ms (2.g) 23dB/10ms

(3.a) 20dB/3.4ms (3.b)20dB/0.25ms(3.c)20dB/0.25ms (3.d) 0dB/1.6ms (3.e)18dB/5.05ms (3.f) 0dB/6.4ms (3.g) 20dB/2.5ms

(4.a) 0dB/4.45ms (4.b) 20dB/1.9ms (4.c) 20dB/0.1ms (4.d) 6dB/6.1ms (4.e) 8dB/0.4ms (4.f) 0dB/5.5ms (4.g)20dB/1.75ms

Fig. 7 Examples captured by optimal exposure settings of the state-of-the-art metrics: **a** Shim [25]; **b** Zhang [29]; **c** Kim [12]; **d** Shin [26]; **e** Tomasi [27]; **f** Wang [28]; **g** Proposed method. The corresponding camera gain and exposure time can be found in the captions. The green dots indicate the matched ORB features ($n_{\max} = 1000$) between stereo image pairs

features. While histogram equalization can enhance contrast, it also tends to amplify noise, potentially decrease the pose estimation accuracy. Therefore, denoising and more complex image preprocessing algorithms are required in this setup, which could be more time/power-consuming.

For better understanding of the comparison to the state-of-the-art methods, some visual examples of both indoor and outdoor scenes are shown in Fig. 7. The analysis is further detailed in Tables 2 and 3. The images are divided into 10×10 blocks, and the total number of matched features (N_{match}) and the number of blocks (N_{block}) with matched feature inside is reported.

As shown in Fig. 7, the proposed method, which optimizes both global and local histograms, consistently produces well-exposed images across a range of scenes. In contrast, other methods may fail to achieve comparable exposure levels under similar scenes. For example for the meeting room, method in Wang [28] prioritizes better exposure of the floor over the desk when the floor enters the field of view. In outdoor scenes with high dynamic ranges, some methods may fluctuate between better exposure of the sky and the foreground.

Table 2 shows that pose estimation using Shi-Tomasi features performs worse for most of the metrics than with ORB features. This is because Shi-Tomasi is more sensitive to noise and intensity variations caused by exposure changes, and methods that maximize gradients or the number of features do not address these constraints. However, the minimum distance between features can be configured in the Shi-Tomasi detection algorithm, which potentially

Table 2 Statistics and pose estimation error corresponding in Fig. 7(1). (Bold: best; Italic: runner-up.)

		a) Shim [25]	b) Zhang [29]	c) Kim [12]	d) Shin [26]	e) Tomasi [27]	f) Wang [28]	g) Proposed
ORB	N_{match}	372	325	*382*	349	**421**	375	366
	N_{block}	*24*	*24*	18	**27**	23	20	21
	E_{t}(cm)	1.5	1.5	**0.5**	*0.7*	2.1	3.1	2.4
	E_{r}(°)	**0.3**	0.6	0.4	**0.3**	*0.4*	*0.4*	*0.4*
Shi-Tomasi	N_{match}	530	**595**	557	100	186	551	*558*
	N_{block}	76	*85*	**86**	36	52	82	84
	E_{t}(cm)	7.8	3.3	*1.6*	9.6	3.2	6.7	**0.4**
	E_{r}(°)	3.0	3.3	2.5	3.8	*1.6*	1.7	**0.9**

Table 3 Statistics and pose estimation error corresponding in Fig. 7(3). (Bold: best; Italic: runner-up.)

		a) Shim [25]	b) Zhang [29]	c) Kim [12]	d) Shin [26]	e) Tomasi [27]	f) Wang [28]	g) Proposed
ORB	N_{match}	*515*	462	462	482	**579**	444	*515*
	N_{block}	*24*	15	15	12	22	21	**26**
	E_{t}(cm)	20.7	*2.3*	*2.3*	21.0	23.3	4.4	**1.6**
	E_{r}(°)	*0.7*	0.8	0.8	1.2	*0.7*	1.2	**0.5**
Shi-Tomasi	N_{match}	619	**710**	**710**	301	686	371	*709*
	N_{block}	39	**62**	**62**	20	47	39	46
	E_{t}(cm)	3.5	25.8	25.8	10.4	7.0	**0.6**	*1.8*
	E_{r}(°)	**1.2**	2.1	2.1	**1.2**	**1.2**	*1.3*	1.4

gives some reliable points from widely distributed features in well-exposed images. For the proposed method, more widespread feature points are detected, as indicated by a larger N_{block}, leading to improved pose estimation accuracy. Furthermore, although the method in [27] maximizes the number of matched ORB features, it does not guarantee the maximization of Shi-Tomasi features.

In the outdoor scene with high dynamic range (Table 3), some methods prioritize exposing the sky. With FAST keypoint detection, most of the ORB features are extracted from the silhouette of the foreground, which can be reliable for pose estimation. In contrast, Shi-Tomasi, giving widespread feature points, may include useless and noisy features from the sky, thereby reducing pose estimation performance, as for Fig. 7(3.b) and (3.c). The proposed metric evaluates local histograms in 3-by-3 blocks and can specifically exclude the histograms from the first row, which are likely to be the sky or ceiling, by adjusting the weights in the loss function. This approach ensures better exposure of the foreground and also benefits other computer vision tasks, such as object detection and semantic segmentation.

In conclusion, the pose estimation performance is determined not only by the number of features, but also by the spatial distribution and quality. The proposed method consistently captures well-exposed images across various scenes, increasing the likelihood of obtaining high-quality features and, consequently, improving pose estimation results.

6.1.2 Noise sensitivity

Signal-to-noise ratio (SNR) is a significant factor for image quality. Noise can enhance existing gradients, which may lead to bias in gradient-based metrics [12, 25, 29]. These metrics use various mapping functions and thresholds to mitigate the impact of noise. In [26], a noise-aware metric is proposed, favoring images with minimal noise by prioritizing lower camera gain. However, since the exposure value EV is the product of the exposure time t and the gain g, in order to maintain the same brightness (exposure value), it requires a longer exposure time, which introduces more motion artifacts. Thus, the noise level and the motion blur should be considered at the same time for exposure strategies.

The proposed exposure control network utilizes a noise-robust metric to optimize the exposure value EV^*. Once the desired exposure value EV^* is set, explicit values for the gain g and the exposure time t need to be selected. The trade-off between noise and motion blur can be balanced by

$$\begin{aligned} &g^*, t^* = \text{argmin}_{g,t} w_{\text{n}} \mathcal{L}_{\text{noise}}(g,t) + w_{\text{m}} \mathcal{L}_{\text{motion}}(t), \\ &\text{s.t.} \quad g \cdot t = EV^*. \end{aligned} \tag{6}$$

$\mathcal{L}_{\text{noise}}$ can be evaluated based on noise model [9]. For $\mathcal{L}_{\text{motion}}$, the motion blur can be calculated as a function of the speed of image pixels approximated by optical flow methods [2, 6].For this gain/exposure time balance strategy, the metric for exposure value optimization should be independent of

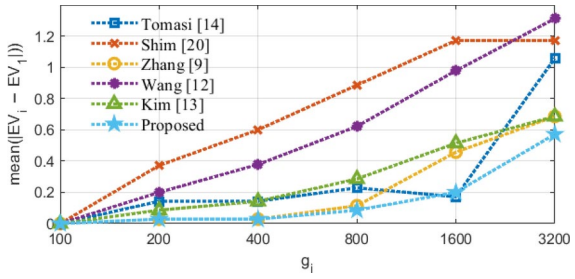


Fig. 8 Noise sensitivity. Smaller value in y-axis indicates more noise robustness

noise. The experiment in this section is designed to evaluate the noise sensitivity of image information metrics. With linear RAW data in the proposed dataset, the images captured by different exposure settings are simulated by the image formation model with noise variances calibrated from Xiaomi 11 Ultra phone.

For the experiment, the *i*-th set of images is simulated with the same camera gain but varying exposure values, as

$$\boldsymbol{S}_i = \{\boldsymbol{I}(g_i, EV) | EV \in [2^{-8}, 2^{-8.5}, ..., 2^{12}]\}. \tag{7}$$

Six image sets are generated with different camera gain, $g_i \in [100, 200, 400, 800, 1600, 3200]$. Given an exposure value, a higher SNR can be achieved with a smaller *g*. Thus, among the six images with the same brightness (exposure value) but different noise characteristics, the image from $\boldsymbol{S}_1$ with $g_1 = 100$ has the highest SNR. For the *i*-th image set, the optimal exposure value is chosen by maximizing the metric *M*,

$$EV_i^* = \text{argmax}_{\{EV|\boldsymbol{I}(g_i,EV)\in\boldsymbol{S}_i\}} M(\boldsymbol{I}(g_i, EV)). \tag{8}$$

Ideally, the same exposure value should be selected as in the absence of noise, which gives small $|EV_i^* - EV_1^*|$ for all $i > 1$.Figure 8 shows $|EV_i^* - EV_1^*|$ averaged over 35 scenes. The absolute difference remaining zero for different *g* indicates that the metric is independent of noise. The result shows that the proposed metric is most robust towards noise compared to the state-of-the-art metrics [12, 25, 27–29].

6.1.3 Ablation study on metric design

The proposed metric is designed based on the similarity between the image histogram and the uniform histogram, and there are various metrics to evaluate the similarity, including Bhattacharyya distance (BD), Kullback-Liebler divergence (KL), χ^2 statistics (CS) or cosine distance (CD) [17]. Let $\boldsymbol{P}$ be the normalized image histogram with *N* bins, each metric is calculated by the following equations, where the different mapping functions give different penalties on bins with extremely high/low frequencies.

Table 4 Comparison between different distance measures. (Bold: best; Italic: runner-up.)

	LMedS		RANSAC	
	E_r (°)	E_t (m)	E_r (°)	E_t (m)
BD	*1.38*	*0.041*	1.35	0.046
DL	**1.35**	0.052	1.30	**0.012**
CS	1.52	0.057	*1.29*	0.036
CD	*1.38*	**0.030**	**1.28**	*0.024*

$$\begin{aligned}
\text{BD}(\boldsymbol{P}, \boldsymbol{Q}) &= \frac{1}{2}\ln N - \ln\left(\sum_i \sqrt{P_i}\right) \\
\text{DL}(\boldsymbol{P}, \boldsymbol{Q}) &= -\frac{1}{N}\sum_i \log(NP_i) \\
\text{CS}(\boldsymbol{P}, \boldsymbol{Q}) &= N\sum_i (P_i - \frac{1}{N})^2 \\
\text{CD}(\boldsymbol{P}, \boldsymbol{Q}) &= 1 - \frac{1}{\sqrt{N\sum_i Pi^2}}
\end{aligned} \tag{9}$$

In order to design the effective metric, the pose estimation accuracy is evaluated with image pairs selected by different similarity measurement. The negative distance in Eq. 9 are implemented as similarity measurement in the metric calculation (Eq. 4). The rotation and translation errors are reported in Table 4. Since best pose estimation performance can be achieved using CD, cosine similarity is selected for metric calculation in this paper.6.2 Network evaluation
The exposure control network is evaluated based on the proposed dataset. The ground truth of exposure value is generated by maximizing the proposed image information metric in Eq. 4 using brute-force search among $\log_2(EV) \in [-8, 12]$ with 0.5 as the step size. The proposed exposure control network is compared with the state-of-the-art methods [26, 27] in this experiment. The Nelder–Mead (NM) method in [26], as an example of the optimization-based control methods, is implemented to optimize the proposed metric; while the network in [27], as the only network based method to the best of our knowledge, is retrained with the mean squared error against the ground truth.

The reason to choose NM method in [26] to compare is that it is a potential solution for real-time applications with efficient searching strategy and fast convergence. For NM optimization, the loss function at candidate exposure settings needs to be evaluated during each iteration. This parameter space sampling can be done by multiple image acquisition or simulation based on reconstructed scene radiant power. For fair comparison and consideration of real-time video capturing, the second method is implemented. A preview SDR image is first captured at $EV(t-1)$ and then the scene radiant power can be reconstructed by the reverse camera response function. The optimization metric at different exposure settings can be calculated based on the reconstructed radiant and the image formation model. At the

end, the optimized exposure value is chosen for image capturing. The processing time of the exposure control includes the time taken for capturing the preview image and the time needed for optimization.

6.2.1 Quantitative comparison

The comparison results are shown in Table 5, where the processing time is tested on a NVIDIA GeForce GTX 1080 Ti GPU, and the prediction accuracy is evaluated by the mean square error (MSE) against the ground truth and the minimum matched features across the sequences. The three proposed systems with and without the NM-refined feedback learning (w/ NM and w/o NM), and with recapture process (w/ NM + recapture) are tested.

The comparison shows that the proposed methods provide the best exposure value prediction. The network without feedback learning can be computed in a short processing time of approximated 4 milliseconds. It achieves slightly better results when compared to NM method in [26] which takes 38 milliseconds and needs preview image acquisition. The feedback learning can efficiently improve prediction accuracy. Thus, if camera response function is available and high frame rate (> 60 fps) is not required in the application, the feedback learning should be implemented.

Table 5 Quantitative Comparison between state-of-the-art and proposed methods. (T in the table indicates the time to capture the preview image.)

	NM iteration	Time (ms)	MSE	Min features
Tomasi [27]	0	0.001	2.855	0
Shin [26]	24.57	$37.90 + T$	0.905	60
Proposed (w/o NM)	0	4.14	0.804	160
Proposed (w/ NM)	7.55	14.52	0.655	121
Proposed (w/ NM + recapture)	7.55	$14.30 + T$	0.331	371

Although the network in [27] is time efficient, it does not work well at the exposure value prediction. The occurrence of zero matched features indicates a localization failure. This network aims to extract some useful features from 3 recent images by a lightweight encoder, and is trained by the mean square error against the ground truth. In this setup, the latent space might not be a good representation of the irradiance changes in the scene, which makes this network not capable of generalization. In order to avoid this issue, the proposed system uses the multi-scale histograms to extract irradiance information from the recent images as network inputs, and co-trains the encoder with a histogram prediction branch.



(a) Shin [26] (b) Tomasi [27] (c) w/o NM (d) w/ NM (e) Recapture (f) Ground truth

Fig. 9 figure examples. **a** Shin [26], **b** Tomasi [27], **c** proposed method (w/o NM), **d** proposed method (w/ NM), **e** proposed method (w/ NM + recapture) and **f** ground truth



Fig. 10 Captured images corresponding to 34 – 38 frames in Fig. 11, including **a** Shin [26], **b** Tomasi [27], **c** proposed method (w/o NM), **d** proposed method (w/ NM), **e** proposed method (w/ NM + recapture) and **f** ground truth

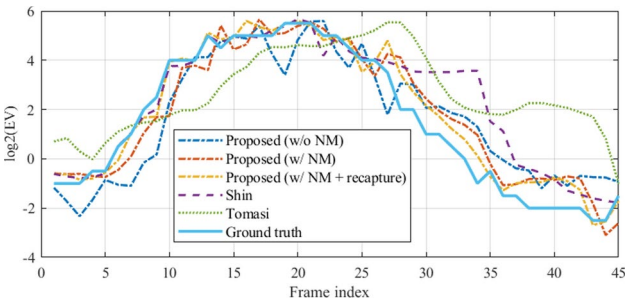


Fig. 11 Comparison on exposure value prediction to the state-of-the-art methods [26, 27] in an indoor–outdoor transition scene with large illumination changes

6.2.2 Qualitative results

Figure 9 shows some examples for figure comparison. Figure 9c and d are, respectively, the result of the proposed methods without and with the NM-refined feedback learning. The comparison shows that the feedback learning can improve the exposure control performance. Further improved images can be recaptured as shown in Fig. 9e. This framework can be considered as an optimization-based method with initialization given by the network outputs. Compared to Fig. 9a, the improved initialization yields better exposed images in less processing time with fewer optimization iterations.

Figures 10 and 11 illustrate the system performance in HDR scenes. Figure 11 shows the predicted exposure value sequences, and Fig. 10 shows the captured images for frame 34 – 38, where there are sudden changes in lighting conditions. It demonstrates that the proposed systems can quickly converge to the ground truth value and capture well-exposed images for robust visual-based localization.

The proposed network outperforms the state-of-the-art exposure control network in [27], as shown in the comparison between Fig. 10b and c. Figure 10d shows that the feedback learning improves the system performance.

Normally NM optimization in [26] can achieve high prediction accuracy, except facing fluctuations in illumination. In Fig. 10a, the localization may fail with few matched features between frame 35 and 36. Under this scene with indoor–outdoor transition, the preview image captured by $EV(t-1)$ might have large saturation regions. The algorithm underestimates in exposure adjustment which could greatly improve image information by unsaturating some part of the image, and thus suffers from slow convergence. In contrast, the optimization in the proposed method is run based on the preview images captured with network

Table 6 Comparison of exposure value prediction accuracy in mean square error (MSE) between training with or without joint-learning

MSE	w/o NM	w/ NM
w/o joint-learning	0.854	0.741
with joint-learning	0.804	0.655

predictive exposure values, which allows better exposure control performance as shown in Fig. 10e.

6.2.3 Ablation study on joint learning

In the joint learning framework, the model is trained on multiple tasks, including exposure control and histogram prediction. The 1D CNN encoder are shared across these two tasks, allowing the model to learn better representations of the scene irradiance. The exposure value prediction results with and without the joint learning in training are shown in Table 6. By minimizing the histogram prediction error, the network can better predict the brightness distribution of the next image and adjust the exposure accordingly.

7 Conclusion

In this paper, an exposure control network is proposed for robust visual odometry in challenging lighting environments. The network takes multi-scale histograms of two recently captured images and the corresponding exposure settings as inputs, and predicts the exposure value for the next frame by optimizing a novel image information metric. The metric is proposed based on cosine similarity between image histograms and the uniform histogram in local and global scales. The metric leads to better localization accuracy and noise robustness in the comparison to other state-of-the-art metrics. A feedback learning framework and a joint learning strategy are implemented to improve the performance of the exposure control network. The network can quickly converge and capture well-exposed images in the experiments with a variety of challenging high dynamic range scenes, which can be a solution for real-time video applications. One limitation of this work is that the gain/exposure time balance strategy is implemented after the exposure value decision is made by the network. Future work can explore the trade-off between noise and motion blur in the loss function design.

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Data availability The proposed dataset can be found by https://drive.google.com/drive/folders/1aTjSEuPMKvv19RZQatH6E0kyJoQeUn8I, and the code for model training and testing is available in https://github.com/leejieyu13/HDR-AE

Declarations

Conflict of interest The authors declare that they have no Conflict of interest.

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